

Preliminary design Report

GROUP - 6

Spring 2026

INTRODUCTION

A Container ship is to be designed to carry cargo in 5500+ TEU units from China to East Coast of US. She has to travel at a service speed of 24 knots and has to pass through Panama canal to reduce the number of days of travel. Moreover, we have chosen our base ship as COSCO New York 5089 TEU with IMO number as 9335185.

BASE SHIP DETAILS

Parameter	Symbol / Unit	Value
Length Overall	L_{oa} [m]	294.05
Length Between Perpendiculars	L_{bp} [m]	283.00
Beam (Midship)	B [m]	32.20
Depth	D [m]	21.60
Draught	T [m]	13.50
Block Coefficient	C_b	0.6985
Displacement	Δ [tonnes]	88,369
Deadweight	DWT [tonnes]	68,235
Capacity	TEU	5,089
Service Speed	[kn]	24.3
Froude Number	F_n	0.237
Engine Power	[kW]	41,130

Table 1: Ship Specifications and Design Parameters

DESIGN PARAMETERS OF DESIRED SHIP

Deadweight & Displacement:

Considering the average weight that could be carried in a typical single TEU container to be 12 - 13 tonnes, the cargo deadweight was calculated out to be 68750 tonnes. Now the ratio of cargo deadweight to displacement(Δ) and the total deadweight to displacement (Δ) are, 0.62 and 0.76 respectively.

Description	Symbol	Unit	Value
Cargo Deadweight	DWT_{cargo}	tonnes	68,750.00
Deadweight Tonnage	DWT	tonnes	84,274.19
Displacement (Mass)	Δ	tonnes	110,887.10
Displacement (Volume)	∇	m^3	100,169.01

Table 2: Displacement and Tonnage Data

First Estimation of Dimensions:

As the ship has to pass through Panama canal as Post-Panamax vessel, we have beam and draft restrictions. One must also consider the fact that the container size is fixed. This makes the length estimation to be done step-wise.

$$\frac{\nabla \Delta}{\Delta} = \frac{\nabla l}{l} + \frac{\nabla b}{b} + \frac{\nabla t}{t} \quad (1)$$

Considering all these restrictions, using geometric similarities, Froude number similarities and regression the estimates of dimensions are: The draft to depth ratio is also

Parameter	Symbol	Unit	Value
Length Overall	L_{oa}	m	315.00
Length Between Perpendiculars	L_{bp}	m	303.66
Beam	B	m	34.55
Draught	T	m	14.48
Depth	D	m	20.324
Estimated Freeboard	F_b	m	5.00

Table 3: Principal Dimensions

maintained within the limit.

Form Coefficients

Coefficient	Symbol	Value
Block Coefficient	C_b	0.680
Midship Section Coefficient	C_m	0.982
Prismatic Coefficient	C_p	0.667
Waterplane Area Coefficient	C_{wp}	0.751
Vertical Prismatic Coefficient	C_{vp}	0.905

Table 4: Hull Form Coefficients:

The C_B was assumed to be same as the base ship when the geometric similarity was done. With this C_B a new displacement is found to be 114356.46 tonnes which almost similar to our first assumed displacement. Further the form coefficients were calculated using,

$$C_M = [1 + (1 - C_B)^{0.35}]^{-1} \quad (2)$$

$$C_{wp} = 0.95 C_p + 0.17 (1 - C_p)^{1/3} \quad (3)$$

Power Estimation:

By matching the Admiralty coefficients of both the Base ship and the design ship, we get SHP (Ship Horse Power) of 45530.15 kW.

$$A_c = \frac{\nabla^{2/3} \cdot v^4}{SHP} \quad (4)$$

The shafting efficiency $\eta_s = 0.92$ we get a BHP of 49489.29348 kW.

Lightship:

The lightweight of a ship includes the steel weight, outfitting weight, machinery weight and other weights that include the other in-built accessories. The steel weight was calculated using K. R. Chapman's formula.

$$W_{st} = 0.007 L_{pp}^{1.759} \cdot B^{0.712} \cdot D^{0.374} \quad (5)$$

Machinery Weight is estimated to 5579.27 tonnes. And the Fitting weight is 3452.98 tonnes. These were also calculated using the emperical formulae.

Weight Component	Unit	Value
Steel Weight	tonnes	6,254.44
Machinery Weight	tonnes	5,579.27
Fitting Weight	tonnes	3,452.98
Margin	tonnes	2,000.00
Total	tonnes	17,286.69

Table 5: Light Weight Estimate

Total Displacement:

Description	Unit	Value
1st Displacement Estimate (Δ_1)	tonnes	114,356.47
Deadweight (DWT)	tonnes	84,274.19
Lightship Weight (LWT)	tonnes	17,286.69
Sum (DWT + LWT)	tonnes	101,560.88

Table 6: Displacement Estimates and Weight Breakdown

The second estimate is slightly lower than the first estimate as we did not consider the crew effects. So we can conclude our estimates to be valid till now.

Stability Parameters:

Table 7: Vessel Stability and Hydrostatic Data

Parameter	Symbol	Unit	Value
Vertical Center of Buoyancy	KB	m	7.60
Longitudinal Center of Buoyancy	LCB	m	-0.56
Transverse Metacentric Radius	BM_T	m	6.48
Longitudinal Metacentric Radius	BM_L	m	420.45
Vertical Center of Gravity	KG	m	13.62
Transverse Metacentric Radius (Keel)	KM_T	m	14.08
Longitudinal Metacentric radius	KM_L	m	428.05
Transverse Metacentric Height	GM_T	m	0.46
Longitudinal Metacentric Height	GM_L	m	420.45
Moment to Change Trim 1 cm	$MCTC$	t·m/cm	1386.95

Sanjay Krishna S - 23NA10040

Manish Ramana K - 23NA10021

Mukesh Vardhan L - 23NA10022

Tharun Kumar P - 23NA10033